The spatial mechanism for Indonesia's failed forest conservation policy

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Abstract

Tropical deforestation may account for as much as one tenth of net global greenhouse gas emissions each year.[32, 23] Any viable effort to mitigate the impact of climate change must address tropical deforestation. In an effort to reduce deforestation, the government of Indonesia enacted a moratorium on logging and agricultural concessions in May 2011.[4] Previous research has shown that this policy did not reduce the deforestation rate as intended; and, in fact, deforestation increased after the moratorium was enacted.[25] This empirical study demonstrates that the moratorium may have *caused* an increase in deforestation. The spatial distribution of deforestation was significantly impacted by the conservation policy, displacing clearing activity from one margin of production to another, latent margin of production. The net effect was an increase in the aggregate rate of clearing. The empirical result can be rationalized with basic production theory, highlighting the importance of considering economic incentives when designing conservation policy.

The 2015 United Nations Climate Change Conference prompted a multinational commitment of US$5 billion to reduce carbon emissions from deforestation.[2] The commitment doubles down on previous efforts to mitigate climate change with forest conservation. Previous research has shown, however, that historical commitments have not reduced deforestation as intended. Most notably, in 2010, Norway pledged US$1 billion in aid to Indonesia, conditional on a significant reduction of its deforestation rate.[26] The Indonesian government responded with a moratorium on new permits granted for development in primary forest.[4] Despite the resource commitment, Indonesia "hasn't seen actual progress in reducing deforestation," according to the Norwegian Minister of the Environment in 2016.[31] Indeed, Margono et al. [25] show that the aggregate rate of deforestation *increased* after the moratorium was instituted; and specifically that within the first year of the moratorium, the rates of both lowland and wetland primary forest cover loss exceeded those within the previous decade. This result is corroborated by both reports in the popular press and other scientific studies.[9, 31] While these inquiries report high rates of deforestation, they do not posit an explanation. Rather, they recommend that "questions concerning the moratorium as a driver of increased deforestation are worthy of investigation."[25] We offer an explanation as to why the moratorium failed, based on shifting spatial patterns of deforestation. We proceed to argue that the observed trends in deforestation patterns can be rationalized with basic production theory, and that economic incentives should be a first-order consideration when designing conservation policy.

The Indonesian moratorium was put into effect in May 2011, prohibiting the award of new licenses to clear natural forest. The moratorium was initially set to expire after two years in May 2013, but was repeatedly extended and continues to exist in 2016.[4] The moratorium originally covered 90% of primary forest in Indonesia, although the spatial extent has been revised several times.[26] Notably, licenses that were granted prior to the moratorium were not rescinded; development of natural forest was allowed to continue within existing concessions. Of the existing concessions, approximately 79% of allocated area was undeveloped in 2010, leaving a significant amount of natural forest designated for legal development.[10] In effect, the moratorium constrained development on one margin with alternatives available and abundant. The net effect of the moratorium is dependent on the rate of substitution between the two margins. We show that the substitution toward the latent, undeveloped margin exceeded the substitution away from business as usual development. We use the island of Borneo as a natural experiment to assess the causal impact of the moratorium. Specifically, we leverage the national boundary that bisects Borneo into Indonesia and Malaysia for identifying variation. Indonesia was subject to the policy, whereas Malaysia was not. The differential policy on the same island constitutes a natural experiment, with deforestation patterns in Malaysia serving as a baseline reference. The outcomes of the two groups (Indonesia and Malaysia) are compared across two time periods (before and after the moratorium) to yield a difference-in-differences estimator of the policy impact. The estimator measures the shift in spatial patterns of deforestation when the moratorium was enacted, conditional on economic variables that may confound or mask the effect. We employ a standard parametric regression, followed by a semi-parametric approach as a robustness check.

We utilize data on forest clearing activity from Global Forest Watch, an online platform that tracks deforestation from satellite imagery. The Forest Monitoring for Action (FORMA) reports forest cover loss at 500-meter resolution and at 16-day intervals.[16] These data are openly accessible through a web service, so that the cost of fully replicating this study is low – arguably much lower than most empirical studies. The FORMA data set was selected from the available, remotely sensed data on deforestation because it is updated at a time scale commensurate with economic decision making. There is a trade-off between the spatial and temporal resolution of the data: higher spatial resolution information on deforestation is updated with lower frequency. With FORMA, we have data for each month between January 2008 and May 2012, yielding 218 observations for each of the forested pixels in Borneo. Forested pixels are defined by a Vegetative Continuous Field (VCF) value of 30% or greater in 2000 – a common standard for forest cover loss data based on NASA's Moderate Resolution Image Spectrometer (MODIS) sensor.[13, 17] The top panel in Figure 2 presents the aggregate number of pixels subject to forest clearing activity in Kalimantan, Indonesia and Sarawak, Malaysia – the states that partition Borneo. The vertical lines indicate the timing of the moratorium, when it was announced and when it was subsequently enacted. Note that the rates of both Indonesian and Malaysian clearing increased after the policy treatment; but also that the difference between the two series increased.

Let ω_{it} indicate the aggregate rate of clearing activity in country $i \in \{\texttt{idn}, \texttt{mys}\}$ and time period $t \in \{1, \ldots, 109\}$. The aggregate rate can be partitioned into two types of clearing activity, remote (r_{it}) and peripheral (p_{it}) clearing, such that $\omega_{it} = r_{it} + p_{it}$. Peripheral clearing is defined as clearing activity that occurs on the periphery of previously existing clusters of

cleared land. Remote clearing is forest cover loss is sufficiently far away from previously cleared land, such that it constitutes *new* clusters of cleared land. Figure 1 illustrates the two types of clearing activity. The black pixels indicate clearing that occurred in an previous period, considered an existing cluster of deforestation. The pixels labeled **A**, **B**, and **C** are considered peripheral clearing activity, whereas pixels **D** and **E** are remote clearing activity. The Let $\tilde{r}_{it} = r_{it}/\omega_{it}$ be the proportion of clearing activity in remote clusters for each country and time period. The estimating equations are given by

$$\omega_{it} = \gamma_0 + \gamma_1 m_t + \gamma_2 c_i + \tau_\omega (m_t \cdot c_i) + \epsilon_{it} \quad (1)$$

$$\tilde{r}_{it} = \alpha_0 + \alpha_1 m_t + \alpha_2 c_i + \tau_r (m_t \cdot c_i) + \epsilon_{it} \quad (2)$$

where $c_i \in \{0, 1\}$ is a binary indicator with $c_i = 1$ when $i = \texttt{idn}$, and where $m_t \in \{0, 1\}$ is a binary indicator with $m_t = 1$ when t is after the policy treatment. We allow for two definitions of policy treatment because the moratorium was not immediately enacted when it was announced. The investment decisions of land developers may be impacted by the announcement alone, so we estimate the policy impact separately for the announcement and enactment, or January 2011 and May 2011, respectively. Then $(m_t \cdot c_i) = 1$ when the observation was in Indonesia, post-policy treatment. The coefficient estimates $\hat{\tau}_\omega$ and $\hat{\tau}_r$ indicate the causal effect of the policy on the target variable. A consistent estimator requires that $\epsilon_{it} \sim N(0, 1)$, or that the unobserved variables impacting the target variables are uncorrelated with explanatory variables. One particularly concerning assumption that is implied by this normality condition is the parallel trends assumption. In context, the assumption for a consistent estimator is that the forest clearing response to economic conditions is the same for both Indonesia and Malaysia. The timing and length of response to, say, an abrupt change in global commodity prices may be different across countries. We relax the assumption with nonparametric matching of local trends, both before and after the moratorium. See the **Supplementary Material, Estimation** for further details.

The bottom panel of Figure 2 displays the price of palm oil, which is an economically important commodity in Indonesia.[29] The price of the agricultural products determine the demand, and in turn the derived demand for cleared land – a critical input to agricultural.[14] Commodity prices across the board increased dramatically during this period.[1] Overlaid on the palm oil price, we report the *difference* in remote clearing as a proportion of total clearing between Indonesia and Malaysia for each time period ($\tilde{r}_{t,\texttt{idn}} - \tilde{r}_{t,\texttt{mys}}$ for each t). By inspection, this difference was lower after the moratorium than before the moratorium, relative to commodity prices. This visualization suggests that investment in new clusters was lower than expected in Indonesia after the moratorium. The quantified results are presented in Table 1. Columns (1) and (2) report the parametric $\hat{\tau}_\omega$ estimates for the two definitions of *post-policy* – when the moratorium was announced and when it was enacted. The estimates indicate that $\hat{\tau}_\omega > 0$, suggesting that the policy precipitated an increase in the aggregate rate of forest clearing activity. This result corroborates previous studies. Columns (3) and (4) report the comparable, parametric estimates for $\hat{\tau}_r$ for when the moratorium was enacted and announced. Columns (5) and (6) reestimate the equation with "snapped" time series, relaxing the assumption of strictly

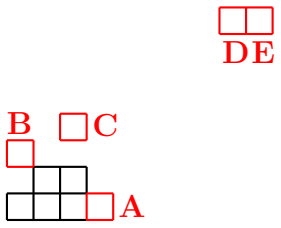


Figure 1: Illustration of clusters

parallel trends near the treatment switch – which serves as a robustness check. Indeed, the estimated treatment effect is reasonably stable with $\hat{\tau}_r < 0$. The proportion of clearing activity in *remote* clusters significantly declined. The results are robust to placebo simulations.[6] In sum, the moratorium increased the aggregate rate of forest clearing activity and consolidated the spatial pattern of clearing activity.

The shift away from remote clearing was more than offset by the shift toward peripheral clearing, precipitating in a higher aggregate rate of clearing activity. The estimated parameters translate into a significant increase in aggregate deforestation, and a significant decrease in the proportion of clearing in peripheral clusters. In response to the moratorium, forest clearing activity in Indonesia became *less* fragmented but *more* intense, conditional on economic variables and relative to Malaysian forest clearing activity.

These results, when taken together, are consistent with basic production theory. A full treatment of the economic theory that underlies agricultural production in Indonesia is beyond the scope of this article, and even beyond the scope of the more extensive structure presented in the **Supplementary Material, Theory**. However, it is worth noting a minimally sufficient set of features that explain the substitution between the two margins of deforestation. Peripheral clearing activity can be conceived as development on the intensive margin, or intensifying clearing on existing clusters. Likewise, remote clearing activity is development on the extensive margin, or the creation of new centers of production. The rate of substitution between the two margins is a function of the relative cost and productivity of the cleared land – a critical input to agricultural or timber production. By definition, the moratorium raised the relative, expected cost of the extensive margin. Development within existing concessions was not *further* restricted by the moratorium. It is natural, then, that producers subject to the new constraint would shift away from extensive clearing at the margin. The net increase in aggregate clearing is consistent with peripheral land as a less productive input than remote land. While this assertion should be examined more extensively (at least to assess the magnitudes) the relation of spatial pattern to revealed prod backed by location theory in agricultural production.[cite] The basic insight is that a plot is seeded by the most productive land, and developed outward until the marginal cost exceeds the marginal benefit.[cite] Together, the differential in productivity and cost increases across margins would yield a net increase in deforestation, given that firms are profit maximizing. It is clear that limiting analysis to one island in the archipelago is not comprehensive; but the profit maximizing, economic structure that is imparted from the theory suggests that the results may be extensible to other areas.

The primary contribution of this paper is to offer empirical evidence that the spatial distribution of deforestation was an important factor in the 2011 moratorium's failure to control deforestation. The results are rationalized by economic theory, which provides a framework to extend the results to future policy design. The primary insight illustrated by the empirical results is that profit-maximizing agents will substitute away from more costly inputs to production. Policy that differentially impacts the costs to inputs will induce a shift toward the relatively cheaper inputs. The profit-maximizing substitution will temper the intended effect of conservation policy. It is important, then, to understand the margins of production when designing policy that is intended to rearrange production. Displacement of deforestation is often a primary policy concern across national borders (often called "leakage") but it is equally

	total clearing (ω_{it})		proportion in peripheral clearing (p_{it})			
	(1)	(2)	(3)	(4)	(5)	(6)
(Intercept)	152.18	374.79***	3.70***	4.15***	4.27***	4.79***
	(140.81)	(125.37)	(0.62)	(0.75)	(0.67)	(0.84)
Indonesia	98.20***	94.32***	2.63***	2.37***	2.62***	2.31***
	(22.96)	(16.79)	(0.10)	(0.10)	(0.11)	(0.11)
post-policy	16.26	75.73***	−0.37***	−0.084	−0.50***	−0.11
	(26.32)	(23.50)	(0.12)	(0.14)	(0.13)	(0.16)
palm oil price	57.58*	3.29	4.41***	3.98**	2.47	2.00
	(34.13)	(30.51)	(1.51)	(1.82)	(1.62)	(2.05)
(palm oil price)^2	−3.09	−0.13	−2.09**	−2.37**	−0.62	−1.02
	(1.98)	(1.77)	(0.88)	(1.06)	(0.94)	(1.19)
policy impact	65.43***	127.62***	−0.84*	−0.54***	−1.04***	−0.72***
	(32.62)	(31.48)	(0.14)	(0.19)	(0.16)	(0.21)
R^2	0.290	0.462	0.834	0.769	0.808	0.708
Adj. R^2	0.274	0.449	0.830	0.763	0.804	0.701
Num. obs.	218	218	218	218	218	218

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table 1: Empirical results.

important to consider other margins across space *and* time.

Supplementary material

Theory

The empirical results are consistent with a basic economic model of production. Weitzman (1976) constructed a model of optimal extraction for a depletable resource from multiple sites. Here, cleared land is treated as a capital input to agriculture, and is stored in the form of primary forest. When primary forest is cleared, the land is activated for agricultural production. Once cleared, the land cannot return to primary forest. Agricultural investment in land can be partitioned into remote and peripheral clearing, depending on its location relative to previously cleared land. The two margins of production have different characteristics, which determine the rate of substitution between the two inputs. The aggregate effect on the input mix when the price of one input changes is a function of the marginal rate of substitution between the inputs. Weitzman's model explicates the conditions that yield higher aggregate deforestation when the expected, marginal cost of one input increases.

Following Weitzman's (2003) notation, define $G(\mathbf{K}, \mathbf{I})$ as the net current cash flow of agriculture, where $\mathbf{K}$ is a vector of capital inputs and $\mathbf{I}$ is a vector of the associated fixed-cost investments. The profit-maximizing developer will choose investment to maximize the present value of $G(\cdot)$ over time. The detailed time-path of development is incidental in determining the aggregate effect of a change in the investment vector. For this type of application, Weitzman

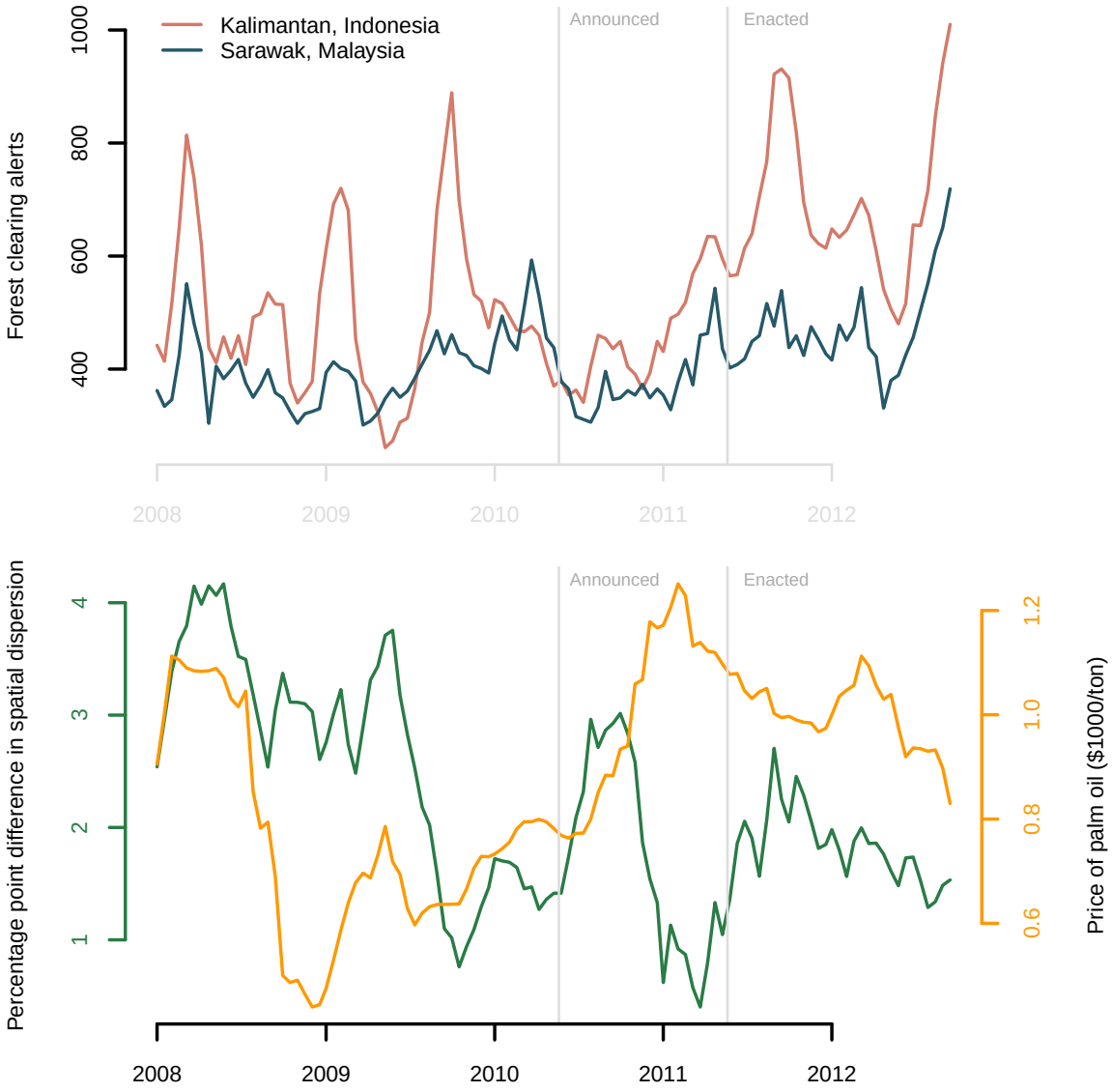


Figure 2: The impact of the moratorium on new forest concessions and commodity prices on the spatial distribution of forest clearing activity.

suggests an "old economist's trick" to collapse the dynamic problem to its stationary equivalent. Consider the prototypical optimal control problem:

$$\max \int_0^\infty e^{-\rho t} G(\mathbf{K}(t), \mathbf{I}(t))\, dt$$
$$\begin{aligned} \text{subject to} \quad & \dot{\mathbf{K}}(t) = \mathbf{I}(t) \\ \text{and} \quad & \mathbf{K}(t) \geq 0 \end{aligned}$$

where $\mathbf{K}(t)$ is the cumulative stock of capital inputs in time t, $\dot{\mathbf{K}}(t)$ is the change in the capital stock in time t, and $\mathbf{I}(t)$ is the instantaneous investment in the corresponding capital inputs. The parameter ρ indicates the competitive interest rate. Define $R(\hat{\mathbf{K}})$ to be the stationary rate of capital return. The stationary solution requires that there exists a time T such that for any $\epsilon_i > 0$ and $t > T$, the optimal solution maintains $\mathbf{I}(t) < \epsilon$. The vector $\hat{\mathbf{K}}$ is the capital input mix that satisfies the conditions for a stationary solution. The stationary rate of capital return is thus defined as

$$R(\hat{\mathbf{K}}) = \frac{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{K}}{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{I}} \tag{3}$$

The fundamental theorem of capital theory sets the stationary rate of return equal to the competitive interest rate, resulting in the the system of equations

$$R(\hat{\mathbf{K}}) = \rho \tag{4}$$

Equation (4) suggests that the investment mix is subject to an external valve, such that the decision to invest in each capital input will be weighed against the going interest rate. Note that the stationary solution may never be reached, but investment decisions will push the capital mix toward the stationary solution over time. An implication of Equation (4) is that, for any two inputs i and j,

$$\frac{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{K}_i}{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{I}_i} = \frac{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{K}_j}{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{I}_j} \qquad \Rightarrow \qquad \frac{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{K}_i}{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{K}_j} = \frac{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{I}_i}{\partial G(\mathbf{K}, \mathbf{0})/\partial \mathbf{I}_j} \tag{5}$$

The system described in Equation (5) demonstrates that at the stationary solution, the bang-per-buck for each input is equal. Otherwise, the profit could be increased by investing more in a different input mix, implying that the solution is not stationary. The implications are not so different from the static, two-factor production model. The present value of the marginal rate of technical substitution should equal the present value of the relative investment costs at the optimum. Two elements of the vector $\hat{\mathbf{K}}$ are cleared land on the periphery of existing clusters and cleared land that would constitute a new, remote cluster. Let $\hat{\mathbf{K}}_1$ be the stationary capital usage for peripheral land, and let $\hat{\mathbf{K}}_2$ be the stationary usage of remote land. These two inputs can be combined to produce a certain level of agricultural product at a competitive market price. The associated revenue, or value to the land developer, is the gross gain Weitzman's $G(\cdot)$ function. At this point, the dynamic problem has been sufficiently collapsed to use the standard insight from a static two-factor production model. The derivation from the dynamic problem ensures that the subsequent insight is robust up to the dynamic considerations faced by the land developer.

The implication in Equation (5) suggests that the optimal solution will remain on a level set of $G(\cdot)$, even after a change in relative input prices. We refer to this level set symbolically as $\bar{G}$, and is depicted in Figure (3). The two inputs, peripheral and remote cleared land, are highly substitutable in agricultural production, such that the level set is almost linear. The rate of substitution, or the slope of the level set, is determined by the marginal productivity of each land type. If peripheral land is, on average, less productive than remote land, then the slope of $\bar{G}$ will be shallow. If peripheral land is relatively more productive, then the slope of $\bar{G}$ will be steep. Note that the slope is not determined by the input price; but rather just the relative productivity of the inputs. Equation (5) provides the link between input price and productivity at the optimum. Specifically, after an input price change, the cost-minimizing input mix will remain on the level set, albeit at a different point.

Suppose that $\hat{\mathbf{K}}_1'$ and $\hat{\mathbf{K}}_2'$ satisfy Equation (5) under an initial cost regime. If the required investment for a unit remote land increases, then the optimum input mix will move along the isoquant to $(\hat{\mathbf{K}}_1'', \hat{\mathbf{K}}_2'')$. This situation corresponds to the moratorium. The moratorium differentially impacted the cost of investment in remote clusters by increasing the uncertainty surrounding the maintenance of the capital input. Given that the moratorium map is uncertain and changes every six-months, the likelihood that a concession granted after May 2011 may be revoked is non-trivial. At best, the moratorium increases the uncertainty of a stranded capital asset (cleared land), and at worst, the moratorium provides leverage to local administrators to extort money from land developers. The rate of corruption surrounding land tenure and development in Indonesia has skyrocketed since the moratorium, according to various local news reports. Either way, the requisite investment for remote clusters increased relative to peripheral clusters as a direct result of the moratorium. Figure (3) indicates that the relative intensity of remote land decreases in response to the price increase.

The effect of the moratorium on the aggregate use of cleared land depends on the average slope of the present value isoquant, which is in turn determined by the relative productivity of the two land types. The dominant use for land cleared at large-scale in Borneo is palm oil. The palm oil production process requires that the raw kernels be processed by a central facility within 24 hours of harvesting. The kernels spoil quickly, and the proportion of spoiled kernels increases in time. The time required to transport the harvested kernels to the processing facility is substantial, given a network of poor, dirt roads. Cleared land that is close to the processing facility therefore has a higher per-acre yield of processed oil than cleared land that is further away. Land on the periphery of existing clusters is, by definition, further away from the seed of the deforestation cluster than the seed itself. New clusters in remote forest landscapes therefore have a higher productivity over the course of the plantation development. Peripheral deforestation indicates that the plantation is further along in its development than remote deforestation, which indicates initial clearing activity. The argument is, in effect, a geometric argument, and reflects the diminishing productivity of a unit of land as the plantation grows.

The characteristics of the two land types support this argument. Note that the tangency of the isocost line would imply that the cost of investment tends to be higher $\hat{\mathbf{K}}_2$ than for $\hat{\mathbf{K}}_1$. It is more difficult to prepare cleared land for agriculture at higher elevations and at higher slope, all else equal. We use the elevation data from the SRTM digital elevation model to examine the characteristics of the two land types. For both Indonesia and Malaysia, the slope and elevation are significantly higher for remote deforestation than for peripheral deforestation (with p-values less than 0.001). This result is consistent with the slope of the isoquant in Figure (3).

Note that, assuming the shallow isoquant in Figure (3), an increase in the cost of investment

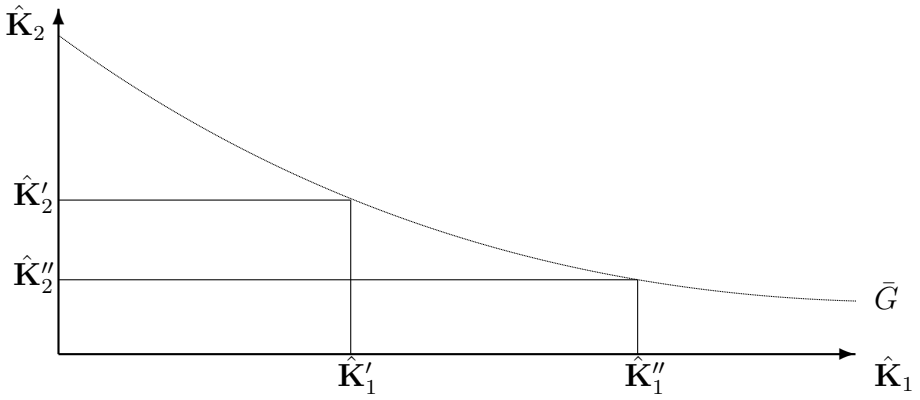


Figure 3: Illustration of an isoquant where the inputs exhibit a high degree of substitution in production and a low marginal rate of technical substitution.

in $\hat{\mathbf{K}}_2$ will yield an increase in the aggregate level of cleared land at the optimum, i.e.,

$$\hat{\mathbf{K}}_1' + \hat{\mathbf{K}}_2' < \hat{\mathbf{K}}_1'' + \hat{\mathbf{K}}_2'' \tag{6}$$

The decrease in $\hat{\mathbf{K}}_2$ is more than offset by the increase in $\hat{\mathbf{K}}_1$ as land developers shift agriculture to the periphery of existing clusters, despite the lower marginal production. After the dynamic investment decisions are collapsed to their stationary equivalents, there is nothing particularly deep about this structure. The empirics indicate that, indeed, more land was cleared in the aggregate after the moratorium, even with a decrease in remote clusters.

Estimation

The estimation strategy is based on the difference-in-difference (DiD) approach, a quasiexperimental method to measure the impact of an abrupt policy change. DiD estimation is common in economics because of its simplicity, but it comes with limitations.[6] An especially concerning limitation in this context is the assumption of parallel paths. The assumption insists that, conditional on the covariates, the potential path in the treatment group is the same as that for the control group. Here, the parallel path assumption requires that (among other things) the patterns in deforestation in the treatment and control groups respond in lock step to exogenous cofactors. It is more likely, however, that producers in Malaysia may respond differently than producers in Indonesia to interest rates, commodity prices, or other economic factors that drive the demand for inputs to agricultural production. We relax the parallel path assumption by employing methods in dynamic time warping (DTW), which are often used in speech processing and gesture recognition. [cite (Gavrila Davis 1995), (Rabiner Juang 1993)]

The basic empirical model is specified in Equations (1) and (2). The target variables, ω_{it} and $\tilde{r}_{it}$, are directly observed from the raw data. There is no processing or smoothing of the measures prior to estimation. The parallel paths assumption applies to these two variables. Consider, however, the top panel of Figure 2. The short-term variation in the aggregate rates of deforestation tend to move together across the two countries. A small bump in deforestation in Indonesia is often accompanied by a small bump in deforestation in Malaysia, albeit shifted or elongated through time (the x-axis). The "small bumps" are clearly induced by an external, possibly unobserved factor. If the external force influences the target variables (a) around the

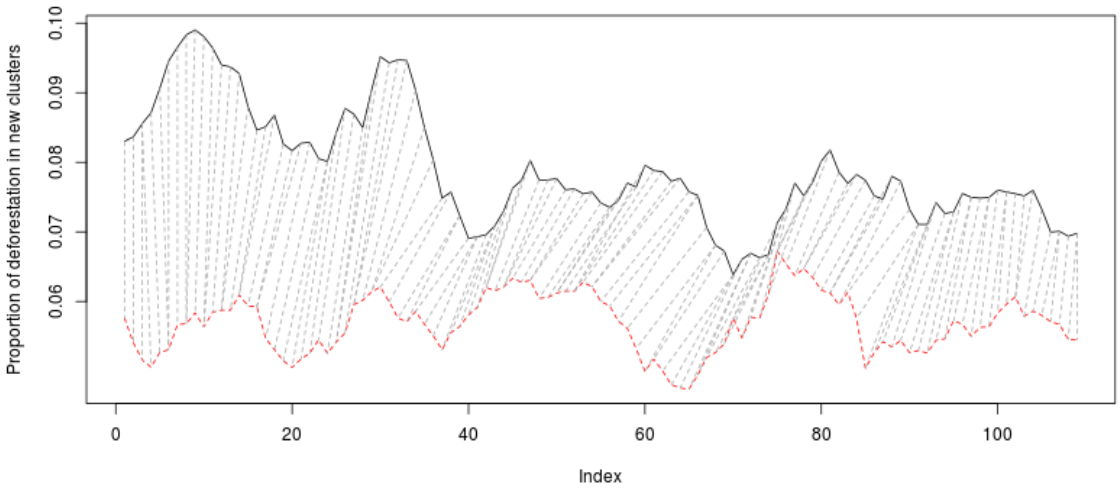


Figure 4: The results of DTW matching.

same time as the policy change, and (b) differentially in the control and treatment groups, then the parallel path assumption may be violated. For example, if the response to an external force is more delayed in Indonesia than in Malaysia, then the effect may be falsely ascribed to the policy – just by virtue of the differential response time.

A possible remedy is to preprocess the target variables to remove transient differences in the series that are not caused by the policy. We are interested in "snapping" the time series for the treatment group to the time series of the control group, based on short-term variation. The objective is to remove any variation induced by heterogenous responses to exogenous factors that may bias the DiD estimator. To this end, we employ DTW methods. DTW effectively warps the time series, searching for the arrangement of the treatment series that minimizes the difference between the treatment and control series, based on short-term variation. The minimization problem is restricted to short-term variation by enforcing a moving window of a given length, although there are many ways to perform dynamic warping. We rely on the `dtw` package in `R`.[] All supporting code is posted online, with assumptions documented. The results of the warping are presented in Figure 4.

The first difference – between treatment and control – is determined by the match shown in Figure 4. The matched series becomes the basis for the second difference – before and after treatment. In some sense, then, the target variables for the treatment group (Indonesia) is preprocessed before the models in Equations (1) and (2) are estimated. Note that DTW will not solve the problem of non-parallel paths that are persistent over time. Rather, it only addresses the edge effects – the potential time shifting close to the time of policy implementation. The DTW procedure should be viewed as a robustness check, alongside the placebo tests also used to check the stability of the results.

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